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Anti-Inflammatory Drugs and Risk of Alzheimer's Disease: An Updated Systematic Review and Meta-Analysis

Jun Wang[a], Lan Tan[a,b,*], Hui-Fu Wang[b], Chen-Chen Tan[a], Xiang-Fei Meng[a], Chong Wang[a], Shao-Wen Tang[c] and Jin-Tai Yu[a,b,d,*]
[a]*Department of Neurology, Qingdao Municipal Hospital, School of Medicine, Qingdao University, Qingdao, China*
[b]*Department of Neurology, Qingdao Municipal Hospital, Nanjing Medical University, Qingdao, China*
[c]*Department of Epidemiology and Biostatistics, School of Public Health, Nanjing Medical University, Nanjing, China*
[d]*Memory and Aging Center, Department of Neurology, University of California, San Francisco, CA, USA*

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Abstract.
Background: In the past 20 years, substantial evidence from laboratory and epidemiologic studies have suggested that anti-inflammatory medications could defer or prevent the occurrence of Alzheimer's disease (AD). However, several studies do not corroborate these findings.
Objective: To evaluate the association of anti-inflammatory drug use on the incidence of AD.
Methods: Pubmed, Embase, and Cochrane Library databases were searched up to March 2014. Studies evaluating the association between use of anti-inflammatory drugs and AD risk were included. Relative risks (RRs) with 95% confidence intervals (CIs) were meta-analyzed using random effects models and were grouped by anti-inflammatory type and duration of drug use.
Results: In observational studies, use of non-steroidal anti-inflammatory drugs (NSAIDs) was significantly associated with a reduced risk of AD (RR, 0.72; 95%CI, 0.62–0.84) compared to no use of NSAIDs, especially in long term users (RR, 0.36; 95%CI, 0.17–0.74); the risks of AD were also lower in both aspirin (RR, 0.77; 95%CI, 0.63–0.95) and non-aspirin NSAID users (RR, 0.65; 95%CI, 0.47–0.88) compared with nonusers; whereas the use of corticosteroids showed no significant association (RR, 0.62; 95%CI, 0.26–1.46). In the single randomized controlled trial (RCT), NSAID use showed no significant effect on AD risk among dementia-free individuals ($p > 0.05$).
Conclusion: Observational studies support the use of NSAIDs for prevention of AD, but RCT do not. Well-designed studies and innovative approaches are required to illuminate the exact relationship between NSAID use and AD risk. The appropriate dosage and duration of use to benefit and the safety are also needed to determine.

Keywords: Alzheimer's disease, anti-inflammatory drug, aspirin, meta-analysis, NSAIDs, prevention, systematic review

INTRODUCTION

Alzheimer's disease (AD) is a complex and multifactorial neurodegenerative disorder and the most common form of dementia in the elderly [1]. It results in a progressive deterioration of intellect, memory, and personality, and creates a great burden for affected individuals, caregivers, and society. Although

*Correspondence to: Dr. Lan Tan or Jin-Tai Yu, Department of Neurology, Qingdao Municipal Hospital, School of Medicine, Qingdao University, No.5 Donghai Middle Road, Qingdao, Shandong Province 266071, China. E-mail: dr.tanlan@163.com. (L. Tan); E-mail: yu-jintai@163.com. (J.T. Yu)

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the pathogenesis of AD is not fully understood, neuroinflammation has been proposed to participate in the pathogenetic cascades of AD [2–5], including the newly proposed pathway involved in defective neuronal insulin signaling [6]. Some pro-inflammatory cytokines related to insulin resistance, like tumor necrosis factor-α (TNF-α) and interleukin (IL)- 1β, can cross the blood-brain barrier and play a role in the crosstalk between peripheral tissues and the brain [7, 8], as has been demonstrated that levels of these pro-inflammatory cytokines were elevated in both AD brains and in peripheral blood [9–11]. Increased brain levels of these pro-inflammatory cytokines can lead to several pathologic features of AD, such as elevated load of oligomeric and fibrillar amyloid-β (Aβ), increased tau phosphorylation, and impaired synaptic plasticity [7], by causing defective insulin signaling. Therefore, it has been recently speculated that chronically elevated peripheral inflammatory mediators may play a role in causing sporadic AD and predispose to earlier onset of AD [6]. So, increasing evidence suggests that anti-inflammatory drugs may serve as a potential preventive or therapeutic approach in AD.

Consistently, in the past 20 years, substantial evidence from laboratory and epidemiologic studies have suggested that anti-inflammatory medications, especially non-steroidal anti-inflammatory drugs (NSAIDs), could defer or prevent the occurrence of AD. However, several studies do not corroborate these findings. In 2003, the meta-analysis of Etminan and colleagues [12], including 9 observational studies, indicated that NSAID use seemed to lower the risk of developing AD. They also reported a potential preventive use of aspirin, but the evidence was not robust. Two other meta-analyses [13, 14] on similar topic also indicated preventive use of NSAIDs in AD incidence, but for aspirin they give conflicting results. For the efficacy of corticosteroids, they did not report. Moreover, several further studies have now been published. Here we conduct an updated and extended systematic review and meta-analysis to evaluate the association of anti-inflammatory drug use on AD incidence and to examine the association by type of anti-inflammatory (e.g., all NSAIDs, aspirin, non-aspirin NSAIDs, and corticosteroids), and duration of use.

METHODS

Search strategy

We conducted this review according to a predefined protocol and in accordance with the PRISMA [15] and MOOSE guidelines [16]. We systematically searched Pubmed, Embase, and the Cochrane library from their commencement to March 2014. The search criteria combined two separate domains: condition (Alzheimer's disease) and intervention (anti-inflammatory drug, NSAID, steroid, corticosteroid, aspirin, cyclooxygenase 2 inhibitor, celecoxib, diclofenac sodium, etodolac, etoricoxib, ibuprofen, indomethacin, ketoprofen, lumiracoxib, meloxicam, naproxen, piroxicam, tenoxicam, rofecoxib, ketorolac, betamethasone, cortisone, dexamethasone, prednisone, or hydrocortisone). Terms were searched in all fields. And the searches were restricted to English-language publications and studies conducted in humans. We also retrieved reference lists of identified articles to find additional potential relevant studies.

Eligibility criteria

Two investigators independently reviewed all pertinent articles using predetermined inclusion criteria. Studies that reported the efficacy of anti-inflammatory drugs on the risk of incident AD among dementia-free individuals were selected for inclusion if they met all of the following criteria: (1) clearly stated the diagnostic criteria for the outcome of AD or identified through diagnostic codes with additional confirmation; (2) explicitly described exposure to anti-inflammatory drugs, defined as any use any time during the study period; (3) presented data on relative risks (RRs) or odds ratios (ORs) with confidence intervals (CIs) or at least presented enough data to allow these to be calculated.

Quality assessment

The quality of included observational studies was appraised with the Newcastle-Ottawa Scale (NOS) criteria [17]. The NOS is an 8-item instrument, and the detail of NOS grading standard was listed as follows: (1) selection, total score: 4; (2) comparability, total score: 2; (3) exposure (case-control studies)/outcome (cohort studies), total score: 3. A high score out of a total of 9 points indicates high methodological quality. As for randomized controlled trials (RCTs), we assessed the methodological quality according to the guidelines of The Cochrane Collaboration's tool for assessing the risk of bias [18]. We assessed studies according to the following criteria: random sequence generation, allocation concealment, blind, incomplete outcome data, selective reporting, and other possible sources of bias.

Data extraction

For each study, two reviewers were blinded to the authors and journal, and independently abstracted data. Discrepancies were discussed and if consensus was not reached, a third reviewer was the final arbitrator. The following data were extracted from each study: study location, year of publication, number of participants, age, outcome definition, exposure definition, variable adjustments, effect estimates and 95% CIs (or information required to compute these), and information required to complete the NOS questionnaire.

When multiple effect estimates were reported, maximally adjusted estimates were extracted. When results were presented with and without lag periods, with multiple lag periods, or with multiple periods of exposure ascertainment, the estimates based on the longest time between exposure and disease onset were used. When more than one studies used the same study population during the same time period, we included only one study with the stronger design (for example, larger sample size, longer follow up, better control of confounding factors).

Statistical analysis

Primary analyses compared exposed with unexposed for each of the four anti-inflammatory exposures of interest: all NSAIDs, aspirin, non-aspirin NSAIDs, and corticosteroids. Moreover, subgroup analyses were conducted to examine differences by duration of use of anti-inflammatory drugs and study design. For all analyses, we used random-effects model with inverse variance method to calculate pooled RRs and 95% CIs as this model accounts for heterogeneity between studies [19–21]. For studies that reported only stratified results, fixed-effect methods with Mantel-Haenszel weighting were used to summarize the stratified estimates into a single parameter for each study. Odds ratios were considered approximations of RRs. Presence of heterogeneity in effects was assessed using the Cochrane Q test and quantified using the I^2 test [22, 23]. A α error $p < 0.20$ and an $I^2 > 50\%$ was regarded as indicators of heterogeneity of outcomes. Publication bias was assessed via visual inspection of the Begg funnel plot. Sensitivity analyses were conducted to assess the robustness of the primary analyses for each exposure type. To explore the potential sources of heterogeneity, a random-effects meta-regression was performed. All statistical tests were two sided and used a significance level of $p < 0.05$. All analyses were performed with STATA12.0 software.

RESULTS

Literature search findings and characteristics of included trails

The search strategy identified 3,956 unique citations. After initial screening based on titles and abstracts, 352 articles remained for further evaluation. Of these articles, 335 were excluded in the subsequent detailed assessments for reasons shown in Fig. 1. One study [24] using overlapping data with another was included because only it provided data on use of aspirin and corticosteroids, while the other one [25] possessed the stronger design. Eventually, 18 studies, including 12 cohort studies [25–36], 5 case-control studies [24, 37–40], and 1 RCT [41] met our inclusion criteria and were included in present systematic review. Among these included studies, 17 reported the association between all NSAID use and AD incidence, 11 reporting specifically the association between aspirin and AD, 8 reporting the association between non-aspirin NSAIDs and AD, and 3 reporting the association between corticosteroids and AD. The characteristics of these studies were listed in Table 1.

Moreover, we assessed the quality of all included studies. All observational studies met 5 NOS criteria, a standard set in other studies [42, 43]. The quality score were listed in Table 1. The single RCT, named The Alzheimer's Disease Anti-inflammatory Prevention Trial (ADAPT), was also good quality. The method of randomization was controlled by coordinating center using permuted blocks and a distributed computerized system. Placebos and anti-inflammatory drugs that were visually identical were used for allocation concealment. Treatments were administered in a

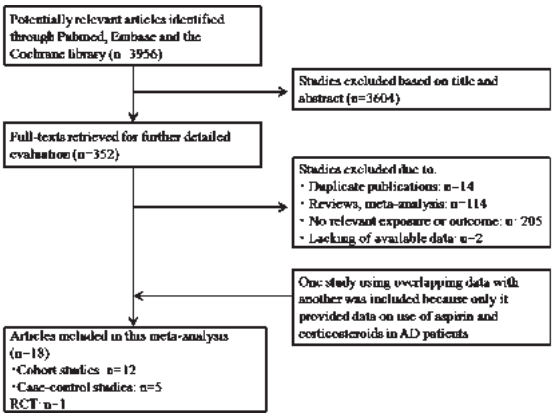

Fig. 1. Flowchart of studies included and excluded in this systematic review.

Table 1
Characteristics of included studies

Study	Location	No.(cases/ controls for case-control studies)	Age (years)	Diagnosis of AD	Variable adjustments	Quality score	Adjusted RR or OR or HR (95%CI)			
							All NSAIDs	Aspirin	Non-aspirin NSAIDs	Corticosteriods
Cohort studies										
Breitner et al., [26]	US	205	NS	NINCDS-ADRDA	Age, sex, acid blockers, insulin	5	0.28(0.14–0.57)	0.34(0.14–0.84)	0.19 (0.06–0.64)	0.57 (0.12–2.63)
Stewart et al. [27]	US	1686	≥65	NINCDS-ADRDA	Age, sex, education, calendar year of follow-up	6	0.64(0.45–0.91)	0.74(0.46–1.18)	0.52 (0.30–0.91)	
in t' Veld et al. [28]	Europe	6989	≥55	NINCDS-ADRDA	age, sex, education, smoking, and concomitant medication	9	0.20(0.05–0.83)			
Zandi et al. [29]	US	3227	≥65	NINCDS-ADRDA	age, sex, education, APOE gene	8	0.75(0.55–1.03)	0.82(0.543–1.23)	0.67 (0.40–1.06)	
Landi et al. [30]	Europe	2708	77.2 (mean)	DSM-IV	age, gender, education, ADL score, CPS score, underlying diseases, and concomitant medication	6	0.43(0.23–0.82)	0.50(0.23–1.10)	0.33 (0.11–0.98)	
Nilsson et al. [31]	Europe	702	≥80	NINCDS-ADRDA	age, sex and various vascular diseases	6		0.41(0.26–0.67)		
Cornelius et al. [32]	Europe	1031	≥75	DSMIII-R	age, sex, education, and underlying diseases	8	1.11 (0.81–1.52)	1.34(0.96–1.89)	0.61 (0.32–1.15)	
Arvanitakis et al. [33]	US	1019	75 (mean)	NINCDS-ADRDA	age, sex, and education	7	0.98 (0.80–1.21)	0.84(0.63–1.11)	1.19 (0.87–1.62)	
Fisher et al. [34]	Europe	585	78.3 (mean)	NINCDS-ADRDA	NS	7	0.40 (0.20–0.81)			
Szekely et al. [35]	US	3229	≥65	NINCDS-ADRDA	age, sex, education, APOE gene, race, baseline 3MSE	7	0.76 (0.61–0.94)	0.87(0.65– 1.16)	0.63 (0.45–0.88)	
Breitner et al. [36]	US	2736	≥65	NINCDS-ADRDA	age, sex, APOE status, education, underlying diseases, body mass index, and regular exercise	8	1.30 (1.03–1.65)			
Côcirc;té et al. [25]	Canada	5276	≥65	NINCDS-ADRDA	Age, sex, education, smoking, alcohol, physical activity, comorbidities, concomitant medication, and vascular risk factors	8	0.73 (0.62–0.87)			

Case-control studies										
Breitner et al. [37]	US	NS	72.5 (mean)	NINCDS-ADRDA	None	5	0.50 (0.10–2.23)	0.56(0.16–1.81)		0.25 (0.06–0.95)
Beard et al. [38]	US	302/302	≥65	NINCDS-ADRDA	Age and sex matched	8	0.78 (0.52–1.15)	0.90(0.54–1.50)	0.79 (0.45–1.38)	
Lindsay et al. [24]	Canada	194/3894	≥65	NINCDS-ADRDA	Age, sex, and education	7	0.65 (0.44–0.95)	0.85(0.55–1.31)	0.62 (0.37–1.04)	1.03 (0.49–2.15)
Yip et al. [39]	US	691/973	≥50	NINCDS-ADRDA	age, sex, ethnicity, education, and APOE gene	7	0.64 (0.38– 1.05)			
Vlad et al. [40]	US	49349/196850	≥50	*International Classification of Diseases*, Ninth Revision, Clinical Modification, code 331.0	age, race, Veterans Affairs priority, comorbidities, and low-dose aspirin use	6	0.76 (0.68–0.85)			
RCT										
ADATP, [41]	US	2257	≥70	NINCDS-ADRDA	Age and clinic		celecoxib versus placebo: 1.03 (0.72–1.50); naproxen versus placebo: 0.92 (0.62–1.35)			

AD, Alzheimer's disease; CI, confidence interval; DSM-IV, Diagnostic and Statistical Manual of Mental Disorders, 4th edition; DSMIII-R, Diagnostic and Statistical Manual of Mental Disorders, revised 3rd edition; MMSE, Mini-Mental State Examination; NS, not stated; NINCDS-ADRDA, National Institute of Neurological and Communicative Disorders and Stroke and the Alzheimer's Disease and Related Disorders Association; RCT, randomized controlled trail; RR, relative risk; OR, odds ratio; HR, hazard ratio.

double-masked fashion. For incomplete outcome data, although there was no mention of an intention-to-treat analysis, the study gave the numbers of dropouts per study arm, and reported that sensitivity analyses incorporating different levels of information about missed outcome events did not change the result meaningfully. ADAPT is the single primary prevention trial, so we have no specific concerns over selective reporting.

All NSAID use and AD risk

Seventeen studies included in our review analyzed the association between all NSAID use and AD risk. Twelve were cohort studies [25–36], 4 were case-control studies [37–40], and 1 was RCT [41]. The pooled RRs of AD with 95%CIs were 0.69 (0.56–0.86) for all NSAID users comparing to nonusers in the cohort studies with a substantial heterogeneity ($I^2 = 79.7\%$), 0.75 (0.68–0.84) in the case-control studies ($I^2 = 0\%$), and 0.72 (0.62–0.84) in overall observational studies ($I^2 = 73.2\%$) (Fig. 2). Sensitivity analyses omitting one study at a time from the original analysis did not change our findings, demonstrating our main results to be robust. Further well-designed and well-conducted studies are still needed to confirm our results. The observational studies revealed that NSAIDs protect against the development of AD, however, the RCT suggested that NSAID use had no significant association with AD risk (celecoxib versus placebo: hazard ratio (HR), 1.03; 95% CI, 0.72–1.50; naproxen versus placebo, HR, 0.92; 95% CI, 0.62–1.35). Begg's test and visual inspection of funnel plots indicated that there was no significant publication bias ($p = 0.099$) (Fig. 3).

Among the 17 studies, five studies [26–29, 40] stratified by the duration of NSAID use, one [37] limited the duration of use to longer than 1 year, and others did not limit or state the duration. Hence, we conducted subgroup analyses according to the duration of drug use (Table 2). In the short term group, four cohort studies and one case-control studies were included. Among these studies, three studies [27–29] limited the duration of use to shorter than 24 months, the RR with 95%CI was 0.79 (0.62–1.01) for short term users of all NSAIDs compared with nonusers. Two studies [26, 40] limited the duration to shorter than 12 months, and the RR with 95%CI was 0.60 (0.14–2.63). The pooled RR with 95%CI of the short term group was 0.89 (0.74–1.07) overall. In the long term subgroup, three studies [27–29] limited the duration to longer than 24 months, while the other three studies [26, 37, 40] limited to longer than 12 months. The results of both of the subgroups in the long term group showed significant protective role of NSAID use on AD incidence, and the effect of subgroup with duration longer than

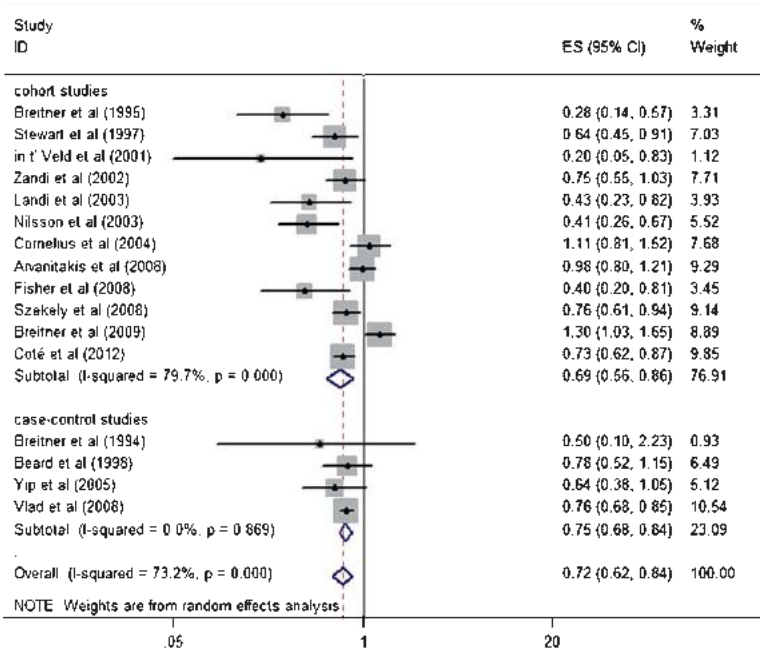

Fig. 2. Forest plot of risk of Alzheimer's disease with all NSAID use. Data type: Effect/CI; Effect Measure: Risk Ratio; Analysis Model: Random Effects; Statistical Method: I-V heterogeneity.

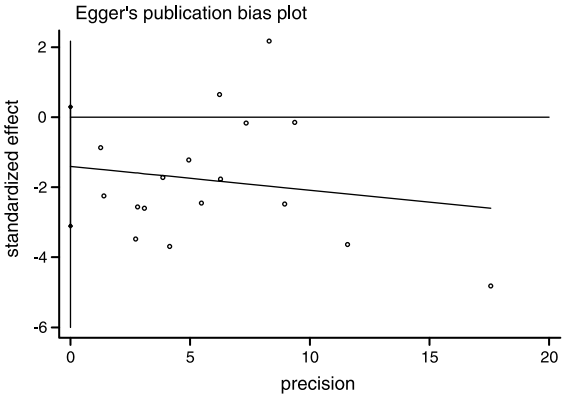

Fig. 3. Funnel plot. Egger's publication bias plot of the use of all non-steroidal anti-inflammatory drugs in Alzheimer's disease risk.

24 months (RR, 0.38; 95%CI, 0.22–0.64) was stronger than that of the other subgroup with duration longer than 12 months (RR, 0.88; 95%CI, 0.85–0.90). The pooled RR with 95%CI was 0.36 (0.17–0.76) overall ($I^2 = 81.2\%$) for long term users of all NSAIDs comparing to nonusers. Our results suggested that long term use, rather than short term use, of NSAIDs significantly protects against AD incidence. Moreover, the effect of continuous long-term use of NSAIDs seemed stronger than the overall cumulative duration of use (RR, 0.72; 95%CI, 0.62–0.84).

The heterogeneity was significant in our meta-analysis ($I^2 = 73.2\%$). Following the subgroup analysis, the heterogeneity remained apparent in the cohort study subgroup ($I^2 = 79.7\%$) and long term subgroup ($I^2 = 81.2\%$). Therefore, to explore the potential sources of heterogeneity, we performed a random-effect meta-regression analysis including all 16 observational studies. The publication year, population origin (U.S. as 1; Europe as 2; and Canada as 3), sample size, the percentage of females, the percentage of low education, quality scores of studies were considered as independent variables respectively (Table 3). The results indicated that the quality scores of studies may be one explanation for the between-study heterogeneity (Fig. 4). In addition, the inclusion criteria for age of participants, dosages of drugs, and adjusted variables are also different between studies, and these may also be the potential sources of the heterogeneity.

Aspirin and non-aspirin NSAID use and AD risk

In our meta-analysis, eight cohort studies [26, 27, 29–33, 35] and three case-control studies [24, 37, 38] explored the association of aspirin use and AD risk. Our results showed that the pooled RR of AD was 0.74 (0.57–0.97) for aspirin users comparing with nonusers in cohort studies ($I^2 = 67.9\%$), 0.84 (0.61–1.16) in case-control studies ($I^2 = 0\%$), and 0.77 (0.63–0.95) in overall ($I^2 = 55.2\%$) (Fig. 5). In addition, seven cohort studies [26, 27, 29, 30, 32, 33, 35] and one case-control study [38] analyzed the impact of non-aspirin NSAID use on incidence of AD. The overall relative risk with 95%CI was 0.65 (0.47–0.88) ($I^2 = 63.0\%$) for non-aspirin NSAID users comparing to nonusers (Fig. 6).

Table 2
Relative risks (95% confidence interval) for all NSAID users, stratified by duration of use

Study	Short term	Long term	Study type
Stewart et al. [28]	0.65 (0.33–1.29) (<24 months)	0.40 (0.19–0.84) (≥24 months)	Cohort study
In't Veld et al. [29]	0.83 (0.62–1.11) (1–23 months)	0.20 (0.05–0.83) (≥24 months)	Cohort study
Zandi et al. [30]	0.75 (0.38–1.34) (≤24 months)	0.45 (0.17–0.97) (>24 months)	Cohort study
Subtotal	0.79 (0.62–1.01)	0.38 (0.22–0.64)	
Breitner et al. [27]	0.188 (0.024–1.491) (1–12 months)	0.075 (0.022–0.261) (>12 months)	Cohort study
Breitner et al. [38]	–	0.50 (0.10–2.23) (>12 months)	Case-control study
Vlad et al. [41]	0.98 (0.95–1.00) (≤12 months)	0.88 (0.86–0.91) (>12 months)	Case-control study
Subtotal	0.60 (0.14–2.63)	0.88 (0.85–0.90)	
Total RRs	0.89 (0.74–1.07)	0.36 (0.17–0.74)	

NSAIDs, non-steroidal anti-inflammatory drugs; -, not available.

Table 3
Results of meta-regression analyses

Variables	Coefficient	Standard error	t	p>|t|	95% confidence interval
Publication year	0.035	0.020	1.80	0.093	–0.007–0.078
Population origin	–0.127	0.164	–0.77	0.453	–0.481–0.226
Sample size	2.15×10^{-6}	8.02×10^{-6}	0.27	0.793	-1.52×10^{-4}–1.95×10^{-4}
Percentage of female	0.001	0.005	0.27	0.795	–0.010–0.013
Percentage of low education	–0.294	0.852	–0.35	0.741	–2.378–1.789
Quality scores	0.239	0.093	2.56	0.023	0.039–0.440

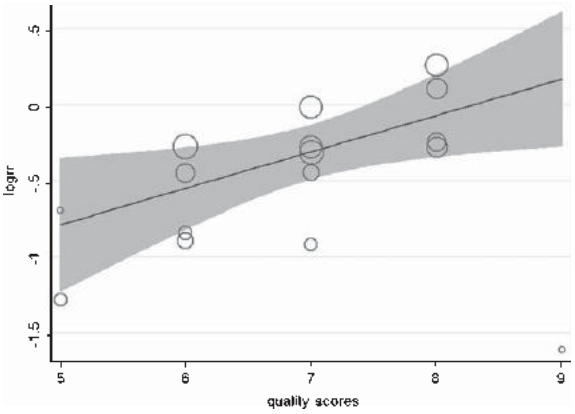

Fig. 4. Meta-regression analysis. The result showed that quality scores of studies may be one explanation of heterogeneity. $p = 0.023$; 95%CI, 0.039–0.440.

Sensitivity analyses omitting one study at a time from the original analysis did not change the overall results, illustrating our findings to be robust. Too few studies were included in these two comparisons; therefore we did not assess the publication bias.

Corticosteroid use and AD risk

Only one cohort study [26] and two case-control [24, 37] studies investigated the relationship between corticosteroid use and incident AD risk. The cohort study [26] (RR, 0.57; 95%CI, 0.12–2.63) and one case-control study [24] (RR, 1.03; 95%CI, 0.49–2.15) did not detect significant association between them, while the other case-control study found that the use of corticosteroids could reduce the risk of incident AD (RR, 0.25; 95%CI, 0.06–0.95) [37]. In our overall meta-analysis, the result showed that there was no significant association between corticosteroid use and AD risk (RR, 0.62; 95%CI, 0.26–1.46) ($I^2 = 38.7\%$).

DISCUSSION

Our results of observational studies, based on analysis of a large number of patients, suggest that use of NSAIDs is associated with a 28% reduction in risk of incident AD. Our results also show a greater benefit with long term rather than short term use (64% reduction). In our meta-analysis, 16 studies reported the association between all NSAIDs and AD risk, 11 studies reported the association between aspirin and AD risk, while 8 studies reported that between non-aspirin NSAIDs and AD risk. All of these three types of NSAIDs show protective effects in AD incidence. For corticosteroids, we failed to detect any significant benefit. In the single RCT included, the results do not support the use of NSAIDs for prevention of AD among dementia-free individuals.

The discrepancies on the effect of NSAID use between observational studies and RCT are especially disconcerting. As mentioned in the review of Szekely

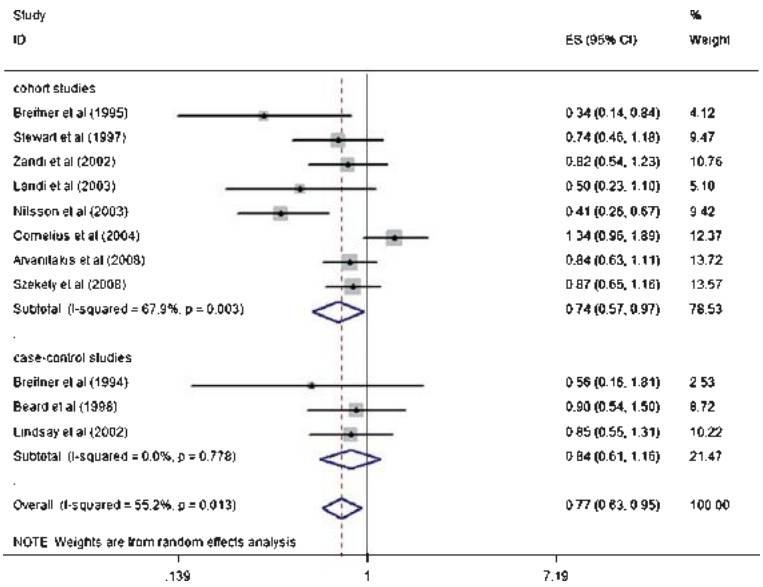

Fig. 5. Forest plot of risk of Alzheimer's disease with aspirin use. Data type: Effect/CI; Effect Measure: Risk Ratio; Analysis Model: Random Effects; Statistical Method: I–V heterogeneity.

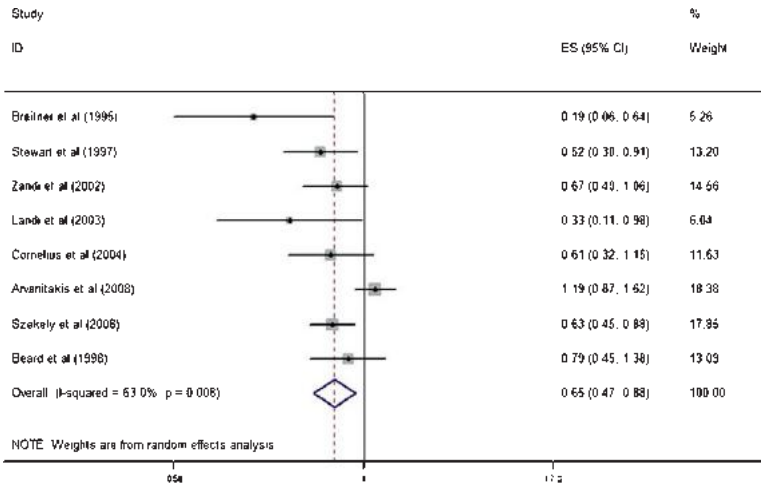

Fig. 6. Forest plot of risk of Alzheimer's disease with non-aspirin NSAID use. Data type: Effect/CI; Effect Measure: Risk Ratio; Analysis Model: Random Effects; Statistical Method: I–V heterogeneity.

et al. [44], some explanations may account for the divergent results. First, the results from observational studies might be spurious due to some types of bias, such as reporting bias and recall bias, especially in some historical or ambispective cohort studies and case-control studies. Besides, confounding by indication, like arthritis, the condition for which NSAIDs are most frequently taken, might also lead to the spurious protective effect. Although several prospective studies have reported that the association between NSAID use and incident AD persisted even after statistically controlling in their analyses for the presence of arthritis, the insufficient sample size and the self-reported arthritis condition with considerable error diminish the ability to tease apart the separate and combined effects of NSAIDs and arthritis on AD. Second, the ADAPT trial is the only available AD primary prevention study randomized clinical trial on NSAIDs. Unfortunately, the ADAPT study was prematurely interrupted for cardiovascular concerns and the median duration of NSAID treatments was far shorter (15 months) than originally intended (7 years). As indicated in our study, NSAIDs may produce a protective effect only if they have been taken a long period, even several years. It is unlikely that this duration of treatment would have sufficed to produce a sustained protective effect in participants. In addition, the ADAPT study was designed hypothesizing an annual incidence of AD of 2.5% during the first year and progressively higher later on. Unfortunately, the actual observed incidence rate over the duration of the trial was only 1.1%. The result was a significant reduction in statistical power below that originally projected. Finally, as suggested [35], NSAIDs might be only effective in the subset of patients with the high risk $\varepsilon 4$ allele at the APOE gene. It is possible that the effect of NSAIDs would be more apparent if the RCT focused particularly on participants with APOE $\varepsilon 4$ allele. Therefore, well-designed studies and innovative epidemiological approaches are required to adequately address the complex problem.

This updated meta-analysis of observational studies, concurring with three previous meta-analyses [12–14], indicates that NSAID use, especially long term use, could significantly prevent the occurrence of AD. However, for aspirin, the results are not the same. One study [12] showed no protective effect of aspirin on AD incidence, while our meta-analysis, as well as another study [13], consistently confirm the role of aspirin in prevention of AD. The eligibility criteria of these meta-analyses are not entirely uniform and therefore the included studies are not identical, which may be the main reason for the different results. Our study has an enhanced power to examine the associations in greater detail. First, we include nine recently published studies, which were not included in previous studies. For studies which used overlapping patient data from the same database, we include only one study with the stronger design. Moreover, we assess the quality of every included observational study, and all meet at least 5 NOS criteria [45]. Furthermore, in addition to NSAIDs, we also analyze the association between corticosteroid use and AD risk. Additionally, sensitivity analyses confirm that our results are robust.

However, several limitations to our meta-analysis restrict the interpretation of our results. First, the analysis of the studies on the use of corticosteroids appears

quite weak being based on only three studies and a limited number of subjects. The point estimates of RR of these three studies were quite low (RR = 0.25, 0.57, and 1.03, respectively); however, the result of the overall meta-analysis indicated no significant association between corticosteroid use and AD risk. This is probably due to limited number of subjects. Incidentally, for one of the studies, the number of subjects is not available. Second, the possibility of confounding and bias may be more significant in meta-analyses of observational studies, and statistical adjustment for confounders in observational studies may not entirely resolve these problems. What is more, the adjustment variables of our included studies are not identical. For example, family history, APOE status, and education, which are important risk factors for AD, were not always adjusted. Moreover, selection bias is more likely in case-control studies, which may unduly weight the outcome in favor of the exposure under evaluation. Furthermore, as discussed above, in some non-prospective cohort studies and case-control studies, information on anti-inflammatory use was obtained by interviewing patients, and recall bias may limit the accuracy of these original studies. Additionally, the restriction of our meta-analysis to English language studies may result in language bias with potentially relevant studies published in other languages being missed.

In short, our findings indicate a protective role of NSAIDs in AD pathogenesis. A possible mode of action for the effectiveness of NSAIDs is by the blockage of COX-2 in the brain. It has been demonstrated that COX-2 mRNA is considerably upregulated in affected areas of AD brain, with COX-2 immunoreactivity located mainly in pyramidal neurons in the cerebral cortex and the hippocampal formation, suggesting the involvement of COX-2 in AD [46, 47]. Besides, NSAIDs may directly affect the production of Aβ [48, 49] by suppressing microglial activation [50] or modifying γ-secretase cleavage away from the more fibrillogenic Aβ_{42} species toward peptides such as Aβ_{40} and Aβ_{42} [48, 51]. In addition, NSAIDs may also function by activating the peraxisomal proliferators-activated receptors, a group of nuclear hormone receptors that act to negatively inhibit the transcription of pro-inflammatory genes, such as IL-6, TNF-α, COX-2, NOS, and cytokines [52, 53].

Whether to recommend the use of inflammatory medicine for preventing AD incidence or not is up to both efficacy and safety. However, due to a limited number of included studies that analyzed safety, we failed to draw a qualitative conclusion whether anti-inflammatory drugs increase the risk of adverse events and mortality in AD prevention, which is one of the problems needed to be solved in future research.

All in all, evidence from our meta-analysis of observational studies indicates that NSAIDs, both aspirin and non-aspirin NSAIDs, may protect against the occurrence of AD, while corticosteroids do not. And long term use of NSAIDs, rather than short term, is associated with a preventive effect. However, the single RCT does not support a preventive effect. Taking into account the divergent results from observational studies and RCT, well-designed studies and innovative approaches are required to resolve the question about the relationship between NSAID use and AD risk. Additional well-designed and well-conducted studies are required to determine the optimal dosage and duration of use and to reliably establish whether anti-inflammatory drugs affect the adverse events and mortality risk in prevention and treatment of AD.

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REFERENCES

[1] Yu JT, Tan L, Hardy J (2014) Apolipoprotein E in Alzheimer's disease: An update. *Annu Rev Neurosci* **37**, 79-100.

[2] Akiyama H, Barger S, Barnum S, Bradt B, Bauer J, Cole GM, Cooper NR, Eikelenboom P, Emmerling M, Fiebich BL, Finch CE, Frautschy S, Griffin WS, Hampel H, Hull M, Landreth G, Lue L, Mrak R, Mackenzie IR, McGeer PL, O'Banion MK, Pachter J, Pasinetti G, Plata-Salaman C, Rogers J, Rydel R, Shen Y, Streit W, Strohmeyer R, Tooyoma I, Van Muiswinkel FL, Veerhuis R, Walker D, Webster S, Wegrzyniak B, Wenk G, Wyss-Coray T (2000) Inflammation and Alzheimer's disease. *Neurobiol Aging* **21**, 383-421.

[3] McGeer EG, McGeer PL (2003) Inflammatory processes in Alzheimer's disease. *Prog Neuropsychopharmacol Biol Psychiatry* **27**, 741-749.

[4] Wang J, Yu JT, Tan MS, Jiang T, Tan L (2013) Epigenetic mechanisms in Alzheimer's disease: Implications for pathogenesis and therapy. *Ageing Res Rev* **12**, 1024-1041.

[5] Tan MS, Yu JT, Jiang T, Zhu XC, Tan L (2013) The NLRP3 inflammasome in Alzheimer's disease. *Mol Neurobiol* **48**, 875-882.

[6] De Felice FG (2013) Alzheimer's disease and insulin resistance: Translating basic science into clinical applications. *J Clin Invest* **123**, 531-539.

[7] Ferreira ST, Clarke JR, Bomfim TR, De Felice FG (2014) Inflammation, defective insulin signaling, and neuronal dysfunction in Alzheimer's disease. *Alzheimers Dement* **10**, S76-S83.

[8] Takeda S, Sato N, Ikimura K, Nishino H, Rakugi H, Morishita R (2013) Increased blood-brain barrier vulnerability to systemic inflammation in an Alzheimer disease mouse model. *Neurobiol Aging* **34**, 2064-2070.

[9] Swardfager W, Lanctot K, Rothenburg L, Wong A, Cappell J, Herrmann N (2010) A meta-analysis of cytokines in Alzheimer's disease. *Biol Psychiatry* **68**, 930-941.

[10] Griffin WS, Stanley LC, Ling C, White L, MacLeod V, Perrot LJ, White CL, 3rd, Araoz C (1989) Brain interleukin 1 and S-100 immunoreactivity are elevated in Down syndrome and Alzheimer disease. *Proc Natl Acad Sci U S A* **86**, 7611-7615.

[11] Grammas P, Ovase R (2001) Inflammatory factors are elevated in brain microvessels in Alzheimer's disease. *Neurobiol Aging* **22**, 837-842.

[12] Etminan M, Gill S, Samii A (2003) Effect of non-steroidal anti-inflammatory drugs on risk of Alzheimer's disease: Systematic review and meta-analysis of observational studies. *BMJ* **327**, 128.

[13] Szekely CA, Green RC, Breitner JC, Ostbye T, Beiser AS, Corrada MM, Dodge HH, Ganguli M, Kawas CH, Kuller LH, Psaty BM, Resnick SM, Wolf PA, Zonderman AB, Welsh-Bohmer KA, Zandi PP (2008) No advantage of A beta 42-lowering NSAIDs for prevention of Alzheimer dementia in six pooled cohort studies. *Neurology* **70**, 2291-2298.

[14] Szekely CA, Thorne JE, Zandi PP, Ek M, Messias E, Breitner JC, Goodman SN (2004) Nonsteroidal anti-inflammatory drugs for the prevention of Alzheimer's disease: A systematic review. *Neuroepidemiology* **23**, 159-169.

[15] Moher D, Liberati A, Tetzlaff J, Altman DG (2009) Preferred reporting items for systematic reviews and meta-analyses: The PRISMA statement. *PLoS Med* **6**, e1000097.

[16] Stroup DF, Berlin JA, Morton SC, Olkin I, Williamson GD, Rennie D, Moher D, Becker BJ, Sipe TA, Thacker SB (2000) Meta-analysis of observational studies in epidemiology: A proposal for reporting. Meta-analysis Of Observational Studies in Epidemiology (MOOSE) group. *JAMA* **283**, 2008-2012.

[17] Wells G Shea B, O'Connell D, Peterson. J, Welch V, Losos M, Tugwell P (Accessed March 4, 2009) The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomised studies in meta-analyses. Ottawa Health Research Institute. Available at: http://www.ohri.ca/programs/clinical_epidemiology/oxford.htm.

[18] Higgins JP, Altman DG, Gotzsche PC, Juni P, Moher D, Oxman AD, Savovic J, Schulz KF, Weeks L, Sterne JA (2011) The Cochrane Collaboration's tool for assessing risk of bias in randomised trials. *BMJ* **343**, d5928.

[19] DerSimonian R, Laird N (1986) Meta-analysis in clinical trials. *Control Clin Trials* **7**, 177-188.

[20] Wang J, Yu JT, Wang HF, Meng XF, Wang C, Tan CC, Tan L. (2014) Pharmacological treatment of neuropsychiatric symptoms in Alzheimer's disease: A systematic review and meta-analysis. *J Neurol Neurosurg Psychiatry,* doi: 10.1136/jnnp-2014-308112.

[21] Wang HF, Yu JT, Tang SW, Jiang T, Tan CC, Meng XF, Wang C, Tan MS, Tan L. (2014) Efficacy and safety of cholinesterase inhibitors and memantine in cognitive impairment in Parkinson's disease, Parkinson's disease dementia, and dementia with Lewy bodies: Systematic review with meta-analysis and trial sequential analysis. *J Neurol Neurosurg Psychiatry,* doi: 10.1136/jnnp-2014-307659.

[22] Higgins JP, Thompson SG (2002) Quantifying heterogeneity in a meta-analysis. *Stat Med* **21**, 1539-1558.

[23] Tan CC, Yu JT, Wang HF, Tan MS, Meng XF, Wang C, Jiang T, Zhu XC, Tan L (2014) Efficacy and safety of donepezil, galantamine, rivastigmine, and memantine for the treatment of Alzheimer's disease: A systematic review and meta-analysis. *J Alzheimers Dis* **41**, 615-631.

[24] Lindsay J, Laurin D, Verreault R, Hebert R, Helliwell B, Hill GB, McDowell I (2002) Risk factors for Alzheimer's disease: A prospective analysis from the Canadian Study of Health and Aging. *Am J Epidemiol* **156**, 445-453.

[25] Côté S, Carmichael PH, Verreault R, Lindsay J, Lefebvre J, Laurin D (2012) Nonsteroidal anti-inflammatory drug use and the risk of cognitive impairment and Alzheimer's disease. *Alzheimers Dement* **8**, 219-226.

[26] Breitner JC, Welsh KA, Helms MJ, Gaskell PC, Gau BA, Roses AD, Pericak-Vance MA, Saunders AM (1995) Delayed onset of Alzheimer's disease with nonsteroidal anti-inflammatory and histamine H2 blocking drugs. *Neurobiol Aging* **16**, 523-530.

[27] Stewart WF, Kawas C, Corrada M, Metter EJ (1997) Risk of Alzheimer's disease and duration of NSAID use. *Neurology* **48**, 626-632.

[28] in t' Veld BA, Ruitenberg A, Hofman A, Launer LJ, van Duijn CM, Stijnen T, Breteler MM, Stricker BH (2001) Nonsteroidal antiinflammatory drugs and the risk of Alzheimer's disease. *N Engl J Med* **345**, 1515-1521.

[29] Zandi PP, Anthony JC, Hayden KM, Mehta K, Mayer L, Breitner JC (2002) Reduced incidence of AD with NSAID but not H2 receptor antagonists: The Cache County Study. *Neurology* **59**, 880-886.

[30] Landi F, Cesari M, Onder G, Russo A, Torre S, Bernabei R (2003) Non-steroidal anti-inflammatory drug (NSAID) use and Alzheimer disease in community-dwelling elderly patients. *Am J Geriatr Psychiatry* **11**, 179-185.

[31] Nilsson SE, Johansson B, Takkinen S, Berg S, Zarit S, McClearn G, Melander A (2003) Does aspirin protect against Alzheimer's dementia? A study in a Swedish population-based sample aged≥80 years. *Eur J Clin Pharmacol* **59**, 313-319.

[32] Cornelius C, Fastbom J, Winblad B, Viitanen M (2004) Aspirin, NSAIDs, risk of dementia, and influence of the apolipoprotein E epsilon 4 allele in an elderly population. *Neuroepidemiology* **23**, 135-143.

[33] Arvanitakis Z, Grodstein F, Bienias JL, Schneider JA, Wilson RS, Kelly JF, Evans DA, Bennett DA (2008) Relation of NSAIDs to incident AD, change in cognitive function, and AD pathology. *Neurology* **70**, 2219-2225.

[34] Fischer P, Zehetmayer S, Jungwirth S, Weissgram S, Krampla W, Hinterberger M, Torma S, Rainer M, Huber K, Hoenigschnabl S, Gelpi E, Bauer K, Leitha T, Bauer P, Tragl KH (2008) Risk factors for Alzheimer dementia in a community-based birth cohort at the age of 75 years. *Dement Geriatr Cogn Disord* **25**, 501-507.

[35] Szekely CA, Breitner JC, Fitzpatrick AL, Rea TD, Psaty BM, Kuller LH, Zandi PP (2008) NSAID use and dementia risk in the Cardiovascular Health Study: Role of APOE and NSAID type. *Neurology* **70**, 17-24.

[36] Breitner JC, Haneuse SJ, Walker R, Dublin S, Crane PK, Gray SL, Larson EB (2009) Risk of dementia and AD with prior exposure to NSAIDs in an elderly community-based cohort. *Neurology* **72**, 1899-1905.

[37] Breitner JC, Gau BA, Welsh KA, Plassman BL, McDonald WM, Helms MJ, Anthony JC (1994) Inverse association of anti-inflammatory treatments and Alzheimer's disease:

Initial results of a co-twin control study. *Neurology* **44**, 227-232.

[38] Beard CM, Waring SC, O'Brien PC, Kurland LT, Kokmen E (1998) Nonsteroidal anti-inflammatory drug use and Alzheimer's disease: A case-control study in Rochester, Minnesota, 1980 through 1984. *Mayo Clin Proc* **73**, 951-955.

[39] Yip AG, Green RC, Huyck M, Cupples LA, Farrer LA (2005) Nonsteroidal anti-inflammatory drug use and Alzheimer's disease risk: The MIRAGE Study. *BMC Geriatr* **5**, 2.

[40] Vlad SC, Miller DR, Kowall NW, Felson DT (2008) Protective effects of NSAIDs on the development of Alzheimer disease. *Neurology* **70**, 1672-1677.

[41] Group ADAPT (2013) Results of a follow-up study to the randomized Alzheimer's Disease Anti-inflammatory Prevention Trial (ADAPT). *Alzheimers Dement* **9**, 714-723.

[42] Millett GA, Flores SA, Marks G, Reed JB, Herbst JH (2008) Circumcision status and risk of HIV and sexually transmitted infections among men who have sex with men: A meta-analysis. *JAMA* **300**, 1674-1684.

[43] Gagne JJ, Power MC (2010) Anti-inflammatory drugs and risk of Parkinson disease: A meta-analysis. *Neurology* **74**, 995-1002.

[44] Szekely CA, Zandi PP (2010) Non-steroidal anti-inflammatory drugs and Alzheimer's disease: The epidemiological evidence. *CNS Neurol Disord Drug Targets* **9**, 132-139.

[45] Wang C, Yu JT, Wang HF, Jiang T, Tan CC, Meng XF, Soares HD, Tan L (2014) Meta-analysis of peripheral blood apolipoprotein E levels in Alzheimer's disease. *PLoS One* **9**, e89041.

[46] Yasojima K, Schwab C, McGeer EG, McGeer PL (1999) Distribution of cyclooxygenase-1 and cyclooxygenase-2 mRNAs and proteins in human brain and peripheral organs. *Brain Res* **830**, 226-236.

[47] Ho L, Pieroni C, Winger D, Purohit DP, Aisen PS, Pasinetti GM (1999) Regional distribution of cyclooxygenase-2 in the hippocampal formation in Alzheimer's disease. *J Neurosci Res* **57**, 295-303.

[48] Weggen S, Eriksen JL, Das P, Sagi SA, Wang R, Pietrzik CU, Findlay KA, Smith TE, Murphy MP, Bulter T, Kang DE, Marquez-Sterling N, Golde TE, Koo EH (2001) A subset of NSAIDs lower amyloidogenic Abeta42 independently of cyclooxygenase activity. *Nature* **414**, 212-216.

[49] Yan Q, Zhang J, Liu H, Babu-Khan S, Vassar R, Biere AL, Citron M, Landreth G (2003) Anti-inflammatory drug therapy alters beta-amyloid processing and deposition in an animal model of Alzheimer's disease. *J Neurosci* **23**, 7504-7509.

[50] Netland EE, Newton JL, Majocha RE, Tate BA (1998) Indomethacin reverses the microglial response to amyloid beta-protein. *Neurobiol Aging* **19**, 201-204.

[51] Eriksen JL, Sagi SA, Smith TE, Weggen S, Das P, McLendon DC, Ozols VV, Jessing KW, Zavitz KH, Koo EH, Golde TE (2003) NSAIDs and enantiomers of flurbiprofen target gamma-secretase and lower Abeta 42 *in vivo*. *J Clin Invest* **112**, 440-449.

[52] Combs CK, Bates P, Karlo JC, Landreth GE (2001) Regulation of beta-amyloid stimulated proinflammatory responses by peroxisome proliferator-activated receptor alpha. *Neurochem Int* **39**, 449-457.

[53] Landreth GE, Heneka MT (2001) Anti-inflammatory actions of peroxisome proliferator-activated receptor gamma agonists in Alzheimer's disease. *Neurobiol Aging* **22**, 937-944.